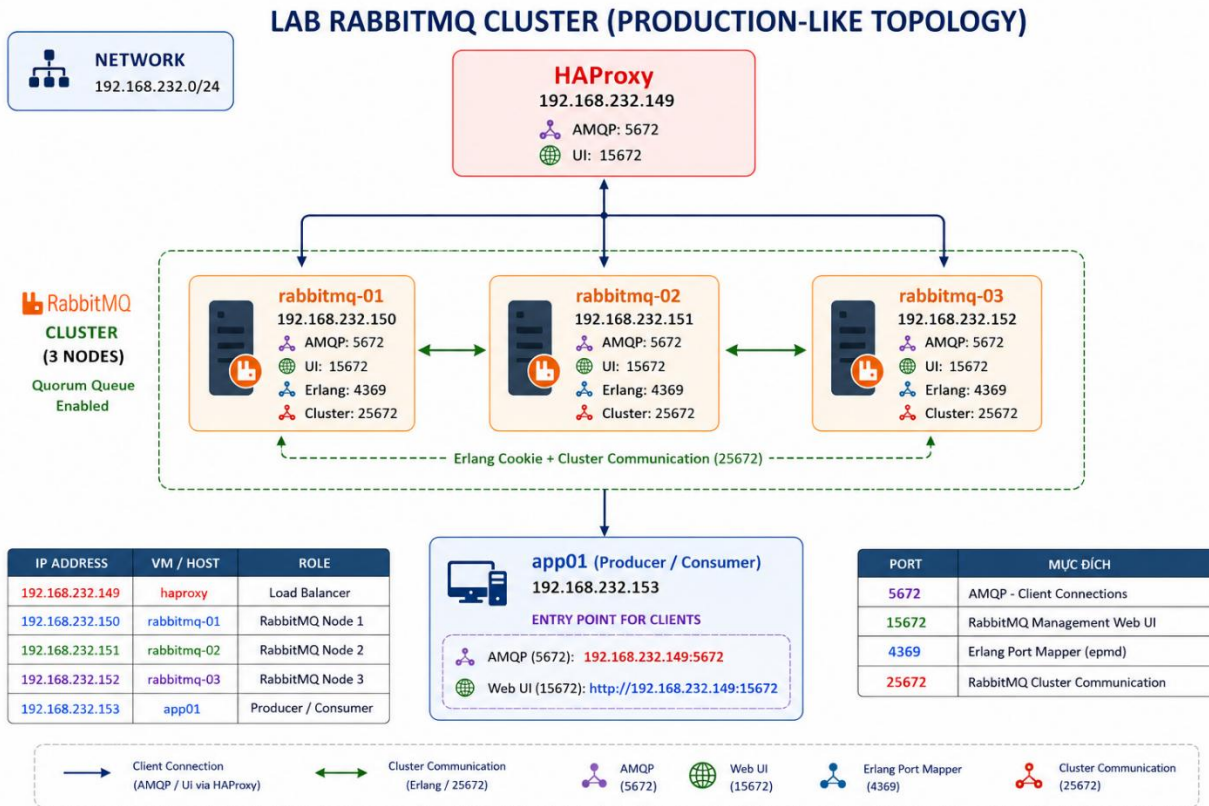


SƠ ĐỒ (TOPOLOGY) LAB RabbitMQ Cluster Production-like



Cấu hình VM hosts:

```
cluster1@rabbitmq-01:~$ sudo cat /etc/hosts
127.0.0.1 localhost
192.168.232.149 haproxy
192.168.232.150 rabbitmq-01
192.168.232.151 rabbitmq-02
192.168.232.152 rabbitmq-03
192.168.232.153 app01
```

Cài RabbitMQ trên 3 node

```

Command python3 from deb python3-is-python3
cluster4@app01:~$ sudo apt install python3-pika -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  python-pika-doc
The following NEW packages will be installed:
  python3-pika
0 upgraded, 1 newly installed, 0 to remove and 32 not upgraded.
Need to get 109 kB of archives.
After this operation, 748 kB of additional disk space will be used.
Get:1 http://vn.archive.ubuntu.com/ubuntu noble/universe amd64 python3-pika all 1.2.0-1 [109 kB]
Fetched 109 kB in 0s (627 kB/s)
Selecting previously unselected package python3-pika.

```

Check port sau khi cấu hình

```

cluster3@rabbitmq-03:~$ curl -I http://192.168.232.152:15672
HTTP/1.1 200 OK
content-length: 3056
content-security-policy: script-src 'self' 'unsafe-eval' 'unsafe-inline'; object-src 'self'
content-type: text/html
date: Wed, 29 Jul 2026 13:31:56 GMT
etag: "1265219175"
last-modified: Thu, 27 Mar 2025 16:08:05 GMT
server: Cowboy
vary: origin

```

Đồng bộ erlang cookie(copy cookie từ node1 sang node2,3)

```

cluster1@rabbitmq-01:~$ sudo cat /var/lib/rabbitmq/.erlang.cookie
[sudo] password for cluster1:
5H2v0+xMAviCUnYIezrGRChFHSOTknTYUB6A0lNXjvDePiJi7dQwJksD

```

```

cluster2@rabbitmq-02:~$ sudo cat /var/lib/rabbitmq/.erlang.cookie
5H2v0+xMAviCUnYIezrGRChFHSOTknTYUB6A0lNXjvDePiJi7dQwJksD

```

Sau khi node2,3 join cluster kiểm tra đồng bộ từ máy 1

```

cluster1@rabbitmq-01:~$ sudo rabbitmqctl cluster_status
Cluster status of node rabbit@rabbitmq-01 ...
Basics

Cluster name: rabbit@rabbitmq-01
Total CPU cores available cluster-wide: 6

Disk Nodes

rabbit@rabbitmq-01
rabbit@rabbitmq-02
rabbit@rabbitmq-03

Running Nodes

rabbit@rabbitmq-01
rabbit@rabbitmq-02
rabbit@rabbitmq-03

Versions

rabbit@rabbitmq-01: RabbitMQ 3.12.1 on Erlang 25.3.2.8

```

Cài HAProxy trên máy app01

```

web@haproxy:~$ sudo journalctl -u haproxy -f
Jul 29 13:34:09 haproxy haproxy[2827]: [WARNING] (2827) : All workers exited. Exiting... (0)
Jul 29 13:34:09 haproxy systemd[1]: haproxy.service: Deactivated successfully.
Jul 29 13:34:09 haproxy systemd[1]: Stopped haproxy.service - HAProxy Load Balancer.
Jul 29 13:34:09 haproxy systemd[1]: Starting haproxy.service - HAProxy Load Balancer...
Jul 29 13:34:09 haproxy haproxy[2860]: [NOTICE] (2860) : haproxy version is 2.8.16-0ubuntu0.24.04.3
Jul 29 13:34:09 haproxy haproxy[2860]: [NOTICE] (2860) : path to executable is /usr/sbin/haproxy
Jul 29 13:34:09 haproxy haproxy[2860]: [WARNING] (2860) : config : parsing [/etc/haproxy/haproxy.cfg:
23] : 'option httplog' not usable with proxy 'rabbitmq_amqp' (needs 'mode http'). Falling back to 'opt
ion tcplog'.
Jul 29 13:34:09 haproxy haproxy[2860]: [NOTICE] (2860) : New worker (2863) forked
Jul 29 13:34:09 haproxy haproxy[2860]: [NOTICE] (2860) : Loading success.
Jul 29 13:34:09 haproxy systemd[1]: Started haproxy.service - HAProxy Load Balancer.

```

Cài thư viện và chạy Producer.py trên app1

```

cluster4@app01:~$ sudo apt install python3-pika -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3-pika is already the newest version (1.2.0-1).
0 upgraded, 0 newly installed, 0 to remove and 32 not upgraded.
cluster4@app01:~$ python3 producer.py
Message sent: {'event': 'student.created', 'student_id': 'S1001', 'name': 'Nguyen Van A', 'grade': 'Gr
ade 10'}

```

Đăng nhập <http://192.168.232.149:15672>

Queues

All queues (3)

Pagination

Page 1 of 1 - Filter: ☐ Regexp ?

Displaying 3 items , page size up to: 100

Overview						Messages			Message rates		
Virtual host	Name	Node	Type	Features	State	Ready	Unacked	Total	incoming	deliver / get	ack
/	school.ad	rabbit@rabbitmq-02 +1	quorum	D Args	running	3	0	3	0.00/s		
/	school.audit	rabbit@rabbitmq-03 +1	quorum	D Args	running	3	0	3	0.00/s		
/	school.email	rabbit@rabbitmq-02 +1	quorum	D Args	running	3	0	3	0.00/s		

Add a new queue

[HTTP API](#) [Documentation](#) [Tutorials](#) [New releases](#) [Commercial edition](#) [Commercial support](#) [Google Group](#) [Discord](#)
[Slack](#) [Plugins](#) [GitHub](#)

HAProxy version 2.8.16-0ubuntu0.24.04.3, released 2026/06/19

Statistics Report for pid 2894

> General process information

pid = 2894 (process #1, nbproc = 1, nbthread = 2)
uptime = 0d 0h 27m 21s; warnings = 3
system limits: memmax = unlimited; ulimitn = 524287
maxsock = 524287; maxconn = 262114; reached = 0; maxpipes = 0
current conns = 2; current pipes = 0/0; conn rate = 0/sec; bit rate = 0.271 kbps
Running tasks: 0/26; idle = 100 %

active UP
active UP, going down
active DOWN, going up
active or backup DOWN
active or backup DOWN for maintenance (MAINT)
active or backup SOFT STOPPED for maintenance

backup UP
backup UP, going down
backup DOWN, going up
not checked

Note: "NOLB"/"DRAIN" = UP with load-balancing disabled.

Display option:

- Scope :
- Hide 'DOWN' servers
- Disable refresh
- Refresh now
- CSV export
- JSON export (schema)

External resources

- Primary site
- Updates (v2)
- Online man

rabbitmq_amqp	Queue		Session rate		Sessions				Bytes		Denied		Errors		Warnings		Server													
	Cur	Max	Limit	Cur	Max	Limit	Cur	Max	Limit	Total	LbTot	Last	In	Out	Req	Resp	Req	Conn	Resp	Retr	Redis	Status	LastChk	Wght	Act	Bck	Chk	Dwn	Dwntm	
Frontend				0	1	-	0	2	262 114	6			2 648	1 780	0	0	0					OPEN								
rabbitmq-01	0	0	-	0	1	0	1	-	2	2	9m59s	1 019	596		0	0	2	0	0	8m34s DOWN	L4CON in 0ms	1/1	Y	-	3	1	8m34s			
rabbitmq-02	0	0	-	0	1	0	1	-	2	2	8m30s	1 045	596		0	0	1	0	0	27m21s UP	L4OK in 0ms	1/1	Y	-	0	0	0	0		
rabbitmq-03	0	0	-	0	1	0	1	-	2	2	8m51s	584	588		0	0	2	0	0	27m21s UP	L4OK in 0ms	1/1	Y	-	0	0	0	0		
Backend	0	0	0	0	1	0	1	26 212	6	6	6m51s	2 648	1 780	0	0	0	5	0	0	27m21s UP		2/2	2	0	0	0	0			

rabbitmq-ui	rabbitmq																													
	Queue			Session rate			Sessions					Bytes		Denied		Errors		Warnings			Status			LastChk		Server				
	Cur	Max	Limit	Cur	Max	Limit	Cur	Max	Limit	Total	LbTot	Last	In	Out	Req	Resp	Req	Conn	Resp	Retr	Redis					Wght	Act	Bck	Chk	Dwn
Frontend				0	6	-	1	7	262 114	18			214 661	1 402 518	0	0	0					OPEN								
rabbitmq-01	0	0	-	0	7	0	2	-	119	116	8m35s	41 380	509 230		0		1	0	3	0	8m34s DOWN	L4CON in 0ms	1/1	Y	-	3				
rabbitmq-02	0	0	-	0	7	0	2	-	227	227	21s	88 006	591 994		0		0	0	0	0	27m21s UP	L7OK/200 in 1ms	1/1	Y	-	0				
rabbitmq-03	0	0	-	0	6	0	2	-	226	226	21s	85 275	301 294		0		0	1	0	0	27m21s UP	L7OK/200 in 1ms	1/1	Y	-	0				
Backend	0	0		0	20	0	6	26 212	589	589	21s	214 661	1 402 518	0	0	1	1	3	0	27m21s UP		2/2	2	0						

	stats																											
	Queue			Session rate			Sessions					Bytes		Denied		Errors		Warnings			Server							
	Cur	Max	Limit	Cur	Max	Limit	Total	LbTot	Last	In	Out	Req	Resp	Req	Conn	Resp	Retr	Redis	Status	LastChk	Wght	Act	Bck	Chk	Dwn	Dwntme	Thrft	
Frontend				0	2	-	1	2	262 114	7		48 746	3 453 597	0	0	0				OPEN								
Backend	0	0		0	2		0	1	26 212	4	0	0s	48 746	3 453 597	0	0	4	0	0	0	27m21s UP		0/0	0	0		0	

```
cluster1@rabbitmq-01:~$ sudo rabbitmqctl cluster_status
```

Listeners

```
Node: rabbit@rabbitmq-02, interface: [::], port: 15672, protocol: http, purpose: HTTP API
Node: rabbit@rabbitmq-02, interface: [::], port: 25672, protocol: clustering, purpose: inter-node and
CLI tool communication
Node: rabbit@rabbitmq-02, interface: [::], port: 5672, protocol: amqp, purpose: AMQP 0-9-1 and AMQP 1.
0
Node: rabbit@rabbitmq-03, interface: [::], port: 15672, protocol: http, purpose: HTTP API
Node: rabbit@rabbitmq-03, interface: [::], port: 25672, protocol: clustering, purpose: inter-node and
CLI tool communication
Node: rabbit@rabbitmq-03, interface: [::], port: 5672, protocol: amqp, purpose: AMQP 0-9-1 and AMQP 1.
0
```

Producer gửi message sau khi node 1 tắt ➔ chứng tỏ HAProxy + RabbitMQ Cluster hoạt động

```
cluster4@app01:~$ python3 producer.py
Message sent: {'event': 'student.created', 'student_id': 'S1001', 'name': 'Nguyen Van A', 'grade': 'Grade 10'}
```

Test producer/consumer

Mở 2 tab VM cho máy app01

Tab 1 chạy poducer/ nhập message

```
cluster4@app01:~$ python3 producer.py
Message sent: {'event': 'student.created', 'student_id': 'S1001', 'name': 'Nguyen Van A', 'grade': 'Grade 10'}
cluster4@app01:~$ |
```

Tab 2 consumer hiển thị / nhận message từ producer

```
cluster4@app01:~$ python3 consumer.py
Waiting for messages...
Received: {'event': 'student.created', 'student_id': 'S1001', 'name': 'Nguyen Van A', 'grade': 'Grade 10'}
```